

AIM

To write a C program to dynamically allocate memory for an array, accept elements from the user, and sort the array using a user-defined function.

PROCEDURE

1. Include the required header files `stdio.h` and `stdlib.h`.
2. Read the size of the array from the user.
3. Allocate memory dynamically for the array using `malloc()`.
4. Read array elements from the user.
5. Create a user-defined function to sort the array.
6. Call the sorting function from the `main()` function.
7. Display the sorted array.
8. Free the dynamically allocated memory using `free()`.
9. Compile and execute the program.

CODE

```
/* Name of the File : dma.c
   Author          : Monish Kanna
   Date           : 17-01-2026
   Purpose        : To create an array using dynamic memory allocation
                   and sort the array using a user-defined function. */

#include <stdlib.h>

void SortArray(int *arr, int n);

int main()
{
    int *arr;
    int n, i;
    printf("Enter number of elements: ");
    scanf("%d", &n);
    arr = (int *)malloc(n * sizeof(int));
    if (arr == NULL)
    {
        printf("Memory allocation failed");
        return 1;
    }
}
```

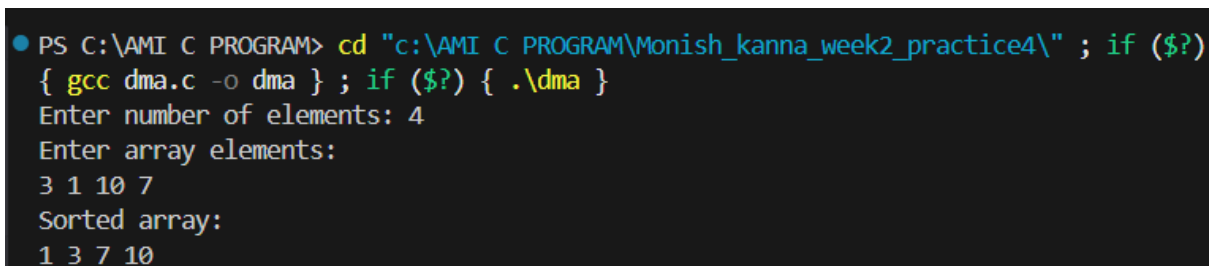
```

printf("Enter array elements:\n");
for (i = 0; i < n; i++)
{ scanf("%d", &arr[i]); }
SortArray(arr, n);
printf("Sorted array:\n");
for (i = 0; i < n; i++)
{ printf("%d ", arr[i]); }
free(arr);
return 0;
}

void SortArray(int *arr, int n){
    int i, j, temp;
    for (i = 0; i < n - 1; i++) {
        for (j = i + 1; j < n; j++) {
            if (arr[i] > arr[j]) {
                temp = arr[i];
                arr[i] = arr[j];
                arr[j] = temp;
            }
        }
    }
}

```

OUTPUT



```

PS C:\AMI C PROGRAM> cd "c:\AMI C PROGRAM\Monish_kanna_week2_practice4\" ; if ($?)
{ gcc dma.c -o dma } ; if ($?) { .\dma }
Enter number of elements: 4
Enter array elements:
3 1 10 7
Sorted array:
1 3 7 10

```

RESULT

Thus, the C program to create an array using dynamic memory allocation and sort it using a user-defined function was successfully executed, and the sorted array was displayed correctly.